Group 35 Final Project Summary Write Up

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Introduction: This report analyzes Chicago crime data from 2025 for public safety officials, urban planners, and residents. The goal is to identify how crime patterns vary by offense type, location, and time so that stakeholders can better understand where and when crime is concentrated. These findings can help support decisions about policing, resource allocation, and community safety planning.

EDA: The exploratory data analysis focused on understanding overall crime patterns in Chicago during 2025 and checking the quality of the dataset before modeling. It included missing-value checks, summary statistics, and grouped summaries for key variables such as crime type, arrest, domestic status, district, and community area to confirm the data was usable and to identify the most common categories. The analysis then examined crime trends by month, hour, and time of day, along with arrest rates across different offense types, to show how crime and arrest outcomes varied over time and by incident category. Visualizations such as bar charts, time-based summaries, and a day-of-week by hour heatmap helped highlight when crimes were most concentrated and which crime types were most associated with arrests. Overall, the EDA established the main descriptive patterns in the dataset and helped motivate the later modeling section by showing that crime outcomes varied meaningfully across offense, time, and location.

Feature Engineering: For the feature engineering portion, we start by taking the date column and separating out new columns that sort for year, month, day, hour, as well as a boolean weekend indicator. From there, we are able to create a condensed column that sorts by 4 different bins based on the time of day, turning a 24 variable column into 4. Beyond this effort, we add an additional categorical column that provides dummy variables based on if the location is described as a street (i.e street or alley) or residential (i.e. residence or apartment). We were able to determine from our data set that Afternoon was the most frequent time of day for crimes and that of our 236,633 responses, 67,558 crimes occurred on the street and 82,514 occurred in a residence. Finally, we were able to detect geographic outliers by determining the center of Chicago as the coordinates of State and Madison and identifying the interquartile range of distances from the center of the city. From this plot, we see that most crime occurred roughly between 2000 and 6000m.

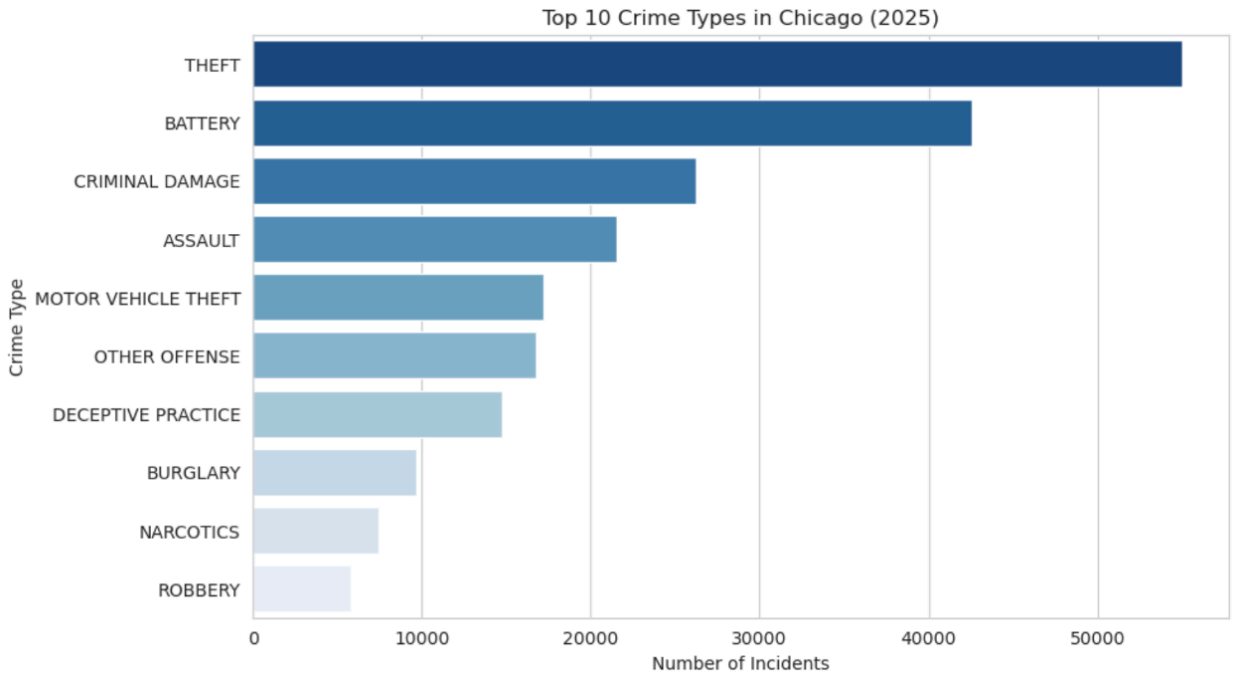
Modeling Analysis: To extend the exploratory analysis, I built two classification models to predict whether a reported crime resulted in an arrest using factors such as crime type, time of day, domestic status, district, and community area. This required additional feature engineering, including extracting date-based variables, grouping high-cardinality categories into more usable groups, and applying one-hot encoding so categorical predictors could be included in the models. The logistic regression model served as an interpretable baseline and achieved an accuracy of

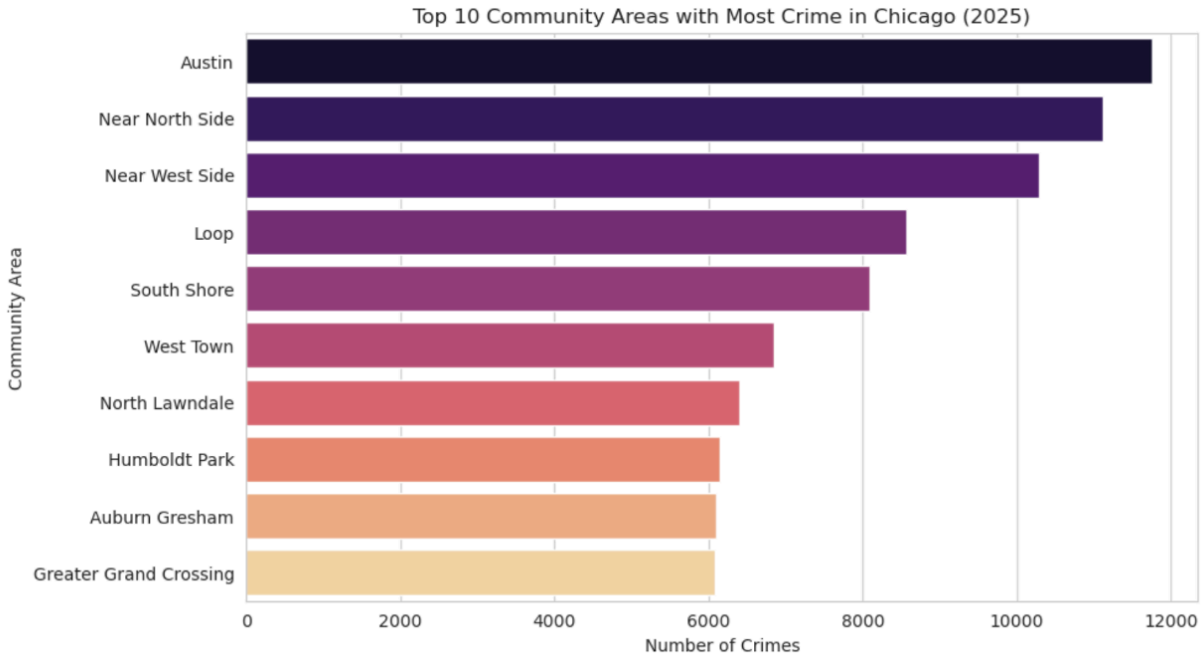
0.8742 and an ROC-AUC of 0.8184, showing good overall separation between arrest and non-arrest cases, though it was relatively conservative in predicting arrests and had low recall for the arrest class. A decision tree model was then fit using the same feature set to capture possible nonlinear relationships and interactions, and it performed slightly better overall, with an accuracy of 0.8843 and an ROC-AUC of 0.8361. The decision tree also improved recall and F1-score for arrest cases, meaning it identified a larger share of true arrests while maintaining similar precision. Overall, both models captured meaningful structure in the data, but the decision tree provided the stronger fit for this classification task because it balanced overall performance with better sensitivity to arrest outcomes.

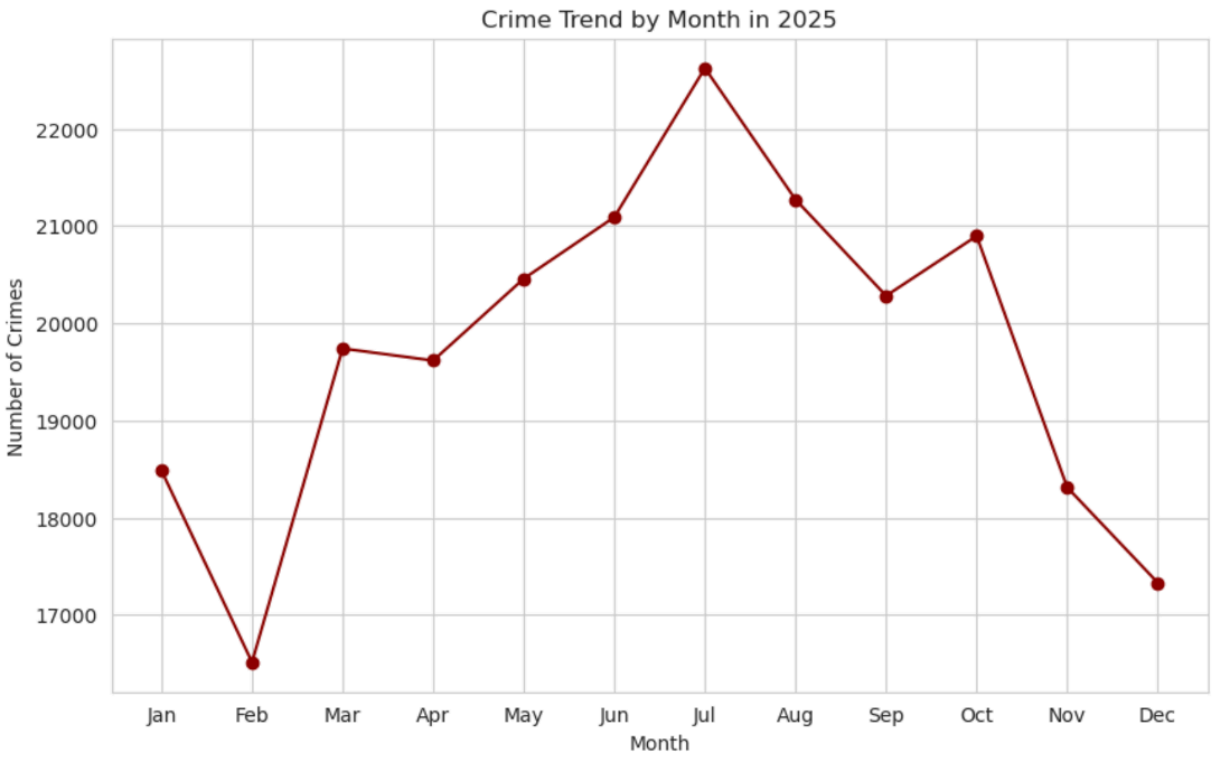
Creativity Section: To incorporate a second dataset, daily Chicago weather data from the Open-Meteo archive API was merged with the crime dataset by date. Days were classified as either rainy or dry based on daily precipitation totals, and daily crime outcomes were then compared across those groups. The results showed that dry days had a slightly higher average daily crime count than rainy days, with about 659 crimes per day on dry days compared with about 637 on rainy days. Average daily arrests followed a similar pattern, and the arrest-rate comparison showed almost no difference between the two weather groups, with both rainy and dry days averaging about a 16% arrest rate. Overall, this suggests that weather may have a modest relationship with daily crime volume, but it does not appear to meaningfully affect the proportion of crimes resulting in arrest.

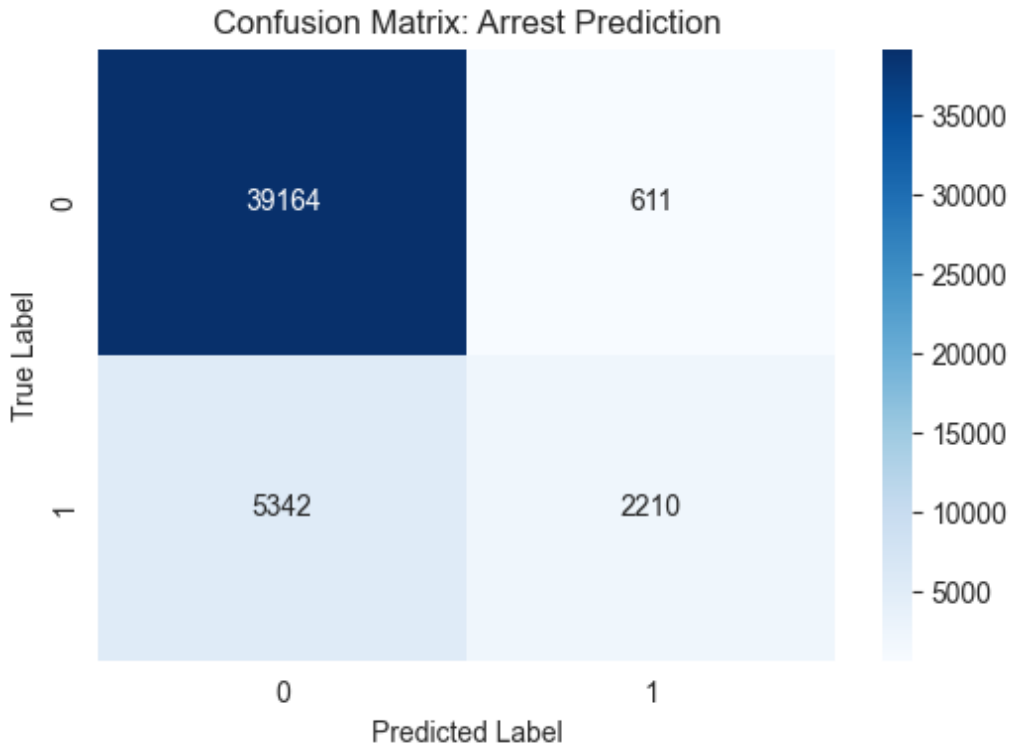
Conclusion: Overall, this analysis showed that Chicago crime patterns in 2025 varied in meaningful ways across offense type, time, and geography, and that these differences were reflected in arrest outcomes as well. The EDA established a strong descriptive foundation by identifying when crimes were most common, which offense categories appeared most frequently, and which types of incidents were more likely to result in arrest. The feature engineering section strengthened the analysis by transforming raw variables into more useful predictors, including date-based features, time-of-day groupings, location-based indicators, and geographic distance measures, which made it easier to detect patterns that would not have been as clear in the original data alone. Building on that, the modeling section showed that these patterns were not only descriptive but also predictive, as both the logistic regression and decision tree models were able to distinguish between arrest and non-arrest cases using crime, time, and location-related features. While both models captured meaningful structure in the data, the decision tree performed slightly better overall and identified a larger share of true arrest cases. Finally, incorporating daily Chicago weather data added a second dataset and provided broader context for the analysis, showing that dry days had slightly higher crime volume than rainy days, while arrest rates remained nearly the same across both conditions. Taken together, these results suggest that arrest outcomes are shaped more strongly by offense, timing, and location context than by simple day-level weather differences, and that these findings can help support more informed discussions about public safety, resource allocation, and crime response planning.

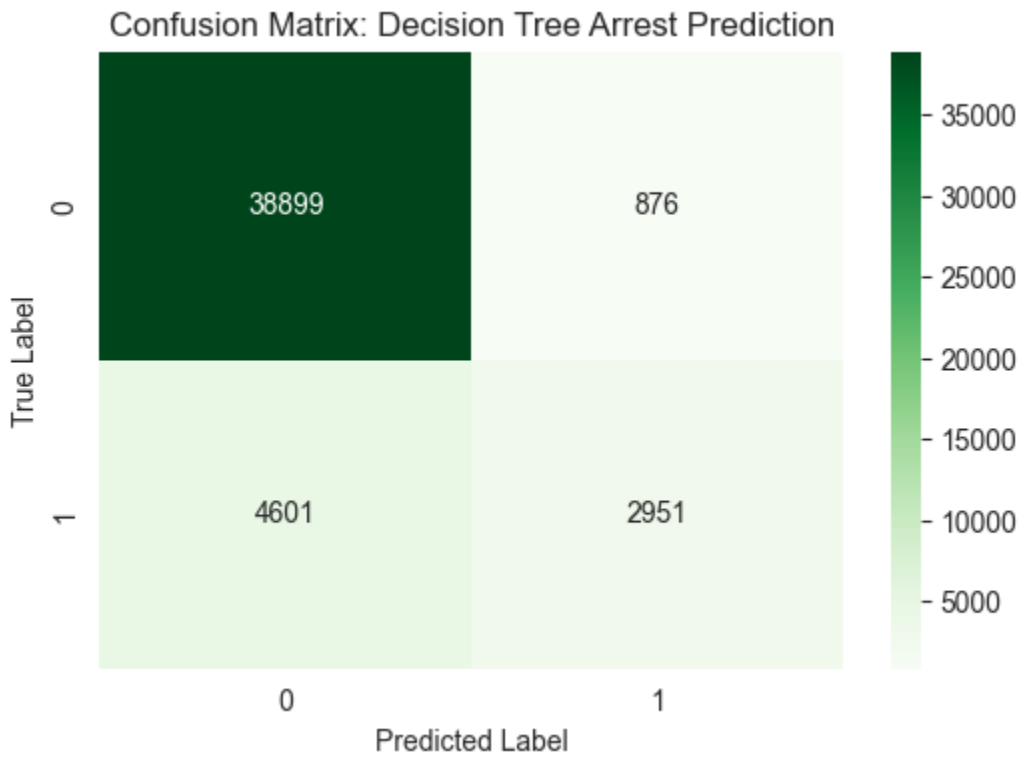
Figure Results:

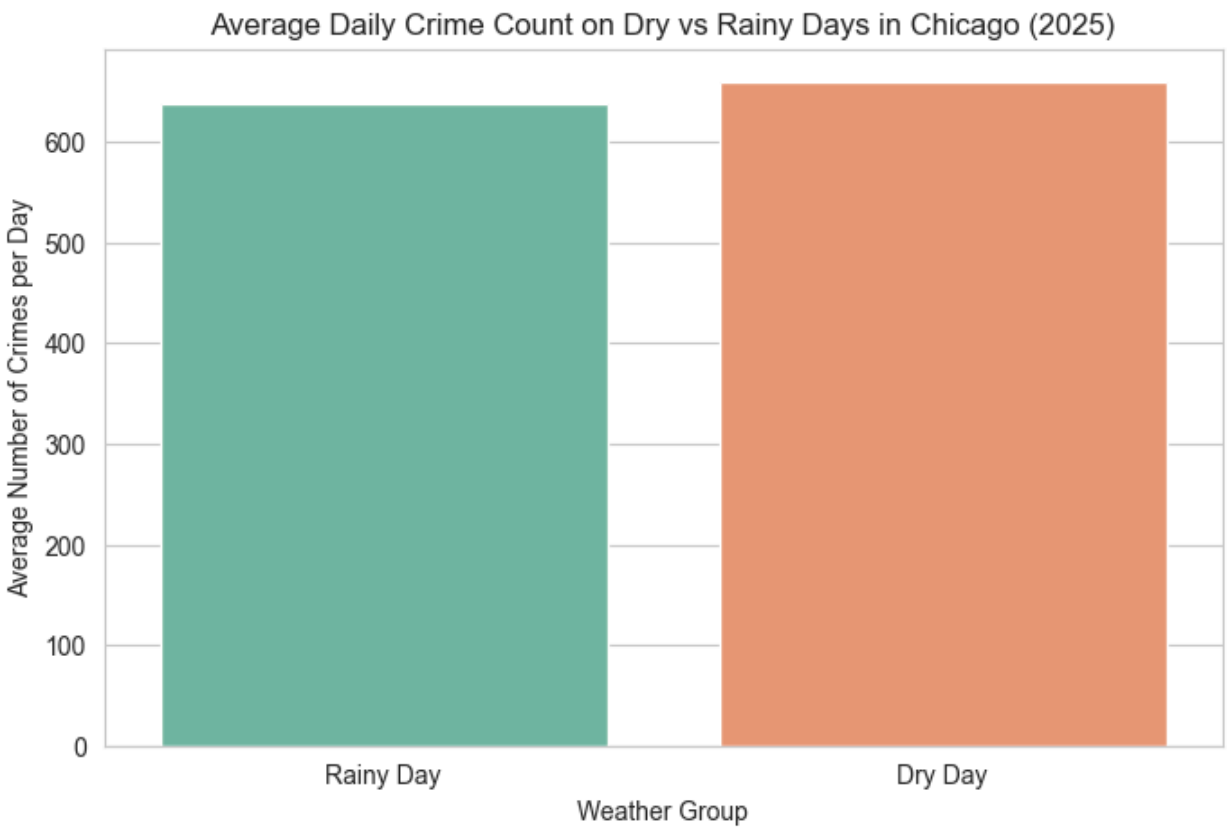


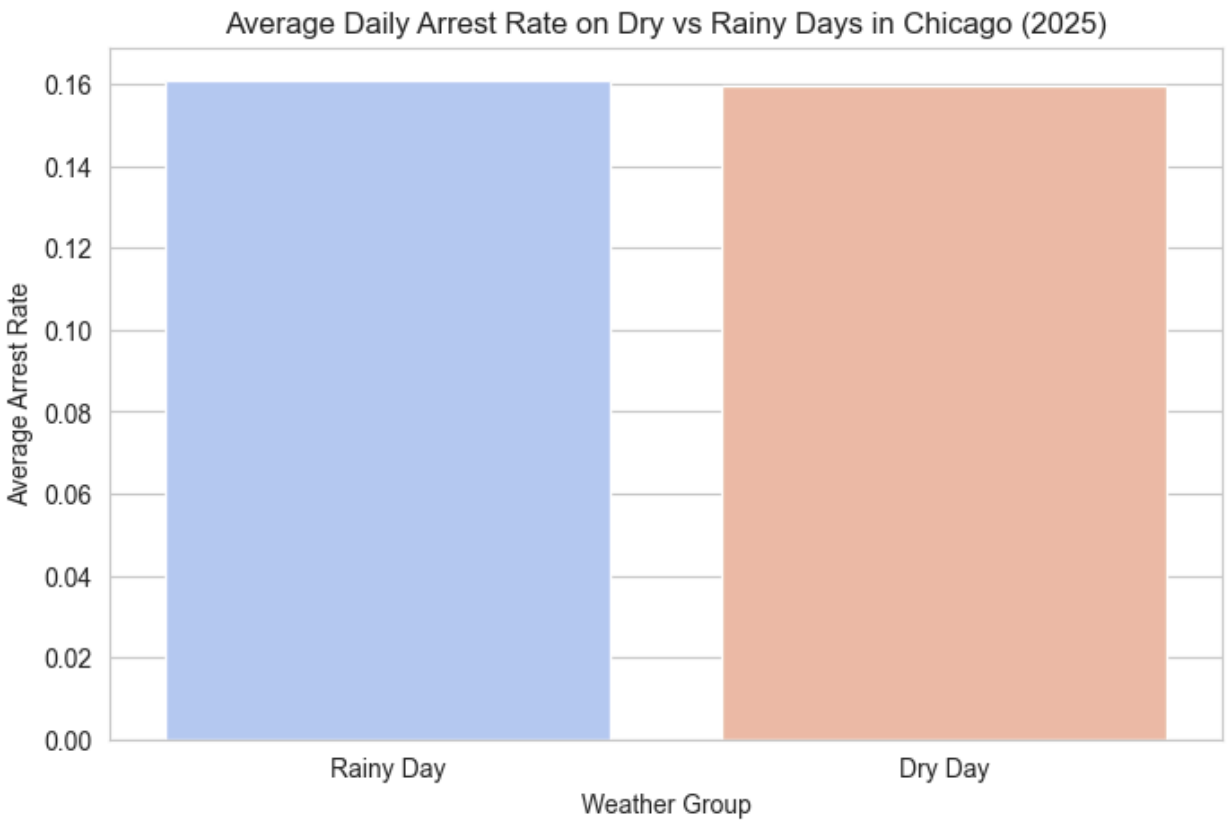


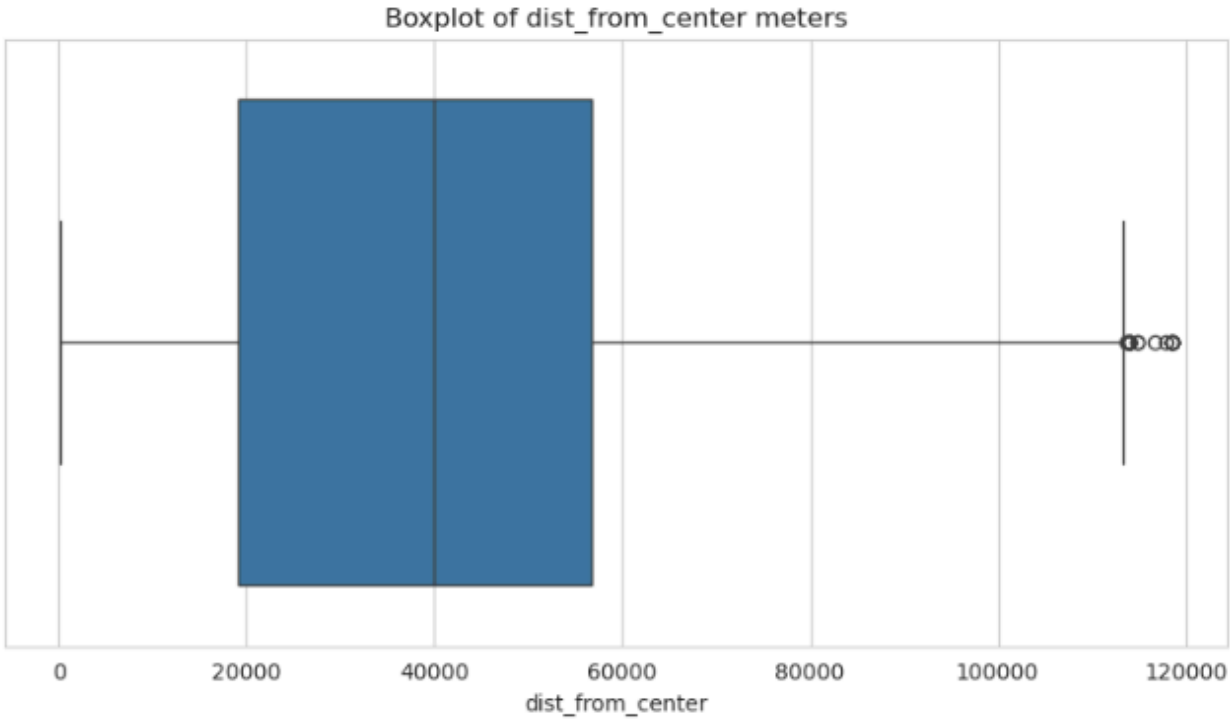

Crime Trend by Month in 2025
Number of Crimes
22000
21000
20000
19000
18000
17000
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month

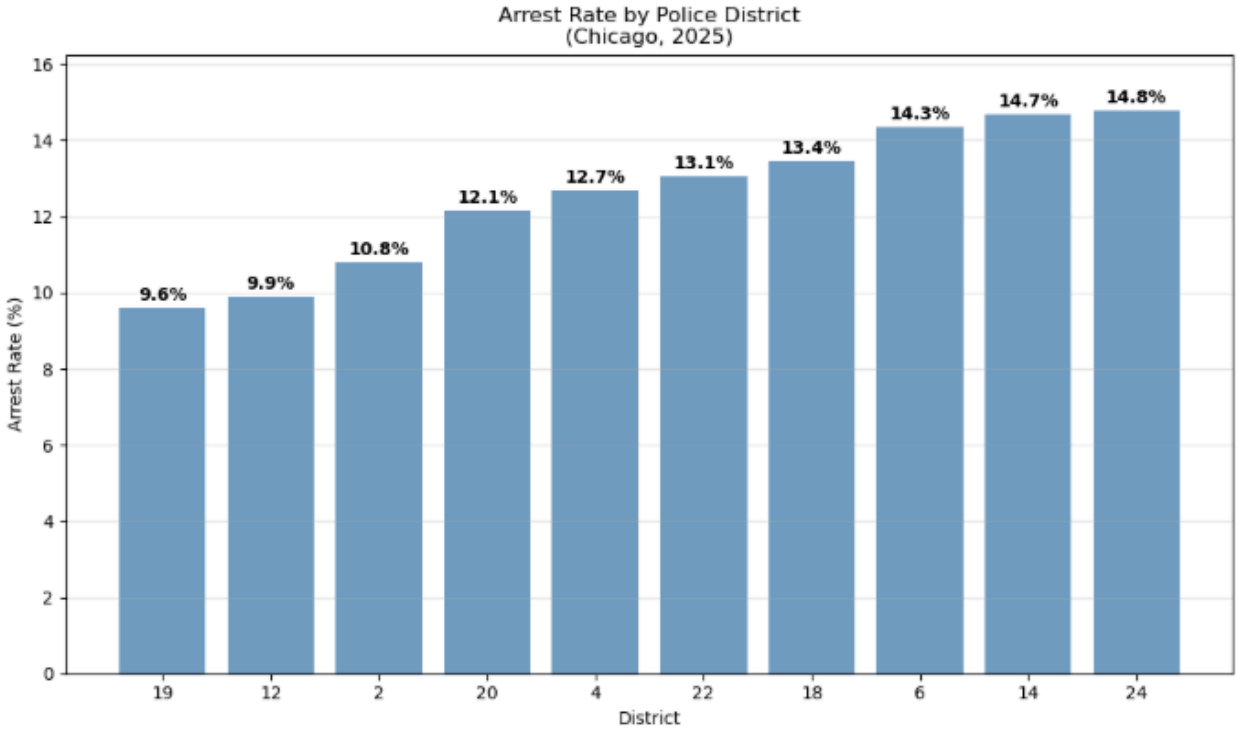

Confusion Matrix: Arrest Prediction
39164
611
5342
2210
True Label
0
1
Predicted Label
0
1
35000
30000
25000
20000
15000
10000
5000


Confusion Matrix: Decision Tree Arrest Prediction
38899
876
4601
2951
True Label
0
1
Predicted Label
0
1
35000
30000
25000
20000
15000
10000
5000


Average Daily Crime Count on Dry vs Rainy Days in Chicago (2025)
Average Number of Crimes per Day
0
100
200
300
400
500
600
Rainy Day
Dry Day
Weather Group


Average Daily Arrest Rate on Dry vs Rainy Days in Chicago (2025)
Average Arrest Rate
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Rainy Day
Dry Day
Weather Group


Boxplot of dist_from_center meters
0
20000
40000
60000
80000
100000
120000
dist_from_center


Arrest Rate by Police District
(Chicago, 2025)
Arrest Rate (%)
0
2
4
6
8
10
12
14
16
9.6%
9.9%
10.8%
12.1%
12.7%
13.1%
13.4%
14.3%
14.7%
14.8%
19
12
2
20
4
22
18
6
14
24
District